

What Drives POWER-CONVERSION COMPONENTS



Priya Ravindran
is a technical
journalist
at EFY

The need of the hour is higher speed at lower power consumption. A laptop is expected to boot-up quickly and its charge to last for at least a few hours. Military equipment must function right in the most stringent conditions. No one wants to go in search of a particular kind of charger for a mobile just because the one he or she has cannot function on the available alternating current (AC) power line in another country.

These are scenarios we never give a second thought to. But essential parts driving each of these are transistors, transformers and rectifiers put together on or around a chip, coming to the fore as a power supply system. Right from a simple torch to something as complex as a satellite system, power-conversion components play a vital role in their functioning. This article tries to look at the latest developments in this sector of electronics and what makes these improvements kick in.

Moving away from silicon

In order to improve system efficiency and lower the form factor, increasing the switching frequency of the system is one viable solution. Critical components for high-frequency action of power-management components are metal oxide semiconductor field-effect transistors (MOSFETs), accompanied by related drivers. Higher the frequency, higher is the required gate charge, and it is common to use MOSFETs in parallel to achieve an

effective lower resistance and higher gate charge. But, there is also a need to make sure that switching loss is lower.

Even with all the enhancements in the silicon wafer, power-conversion components ask for more. While going to lower and lower nodes certainly helps achieve higher switching frequencies in the order of GHz, power-related devices demand a MHz range. Thus, there is a shift towards using materials like gallium-nitride (GaN) and silicon-carbide (SiC).

Lower switching loss. GaN and SiC offer a higher band gap as compared to silicon, and thus the system design itself undergoes a change. GaN MOSFETs were, in the past, mainly used in low-voltage applications (< 100V). Now, GaN based devices target voltage ranges from 200V to 700V, while SiC based devices target a voltage range above 700V. The first 600V-650V GaN was commercialised by Transphorm & Efficient Power Conversion Corp. (EPC) in 2014, and now companies like Infineon and Texas Instruments (TI) have joined this trend. Started off by Cree, SiC MOSFETs are now being manufactured by Microsemi, STMicroelectronics, Toshiba and Infineon, to name a few.

Gate-charge driver is as important.

While the main advantage is that GaN and SiC offer high-frequency support with lower switching loss, and a smaller form factor, driving the gate circuit for these materials requires a negative voltage to be applied, moving away from the traditional silicon based approach. To make it easy to cope with this change, designers are coming up with specialised drivers.

While there have been efforts to provide components and drivers in a single package for easy end use, the design process is complex. Some vendors prefer to separate out the two in order to offer more flexibility in matching existing drivers with transistors of different ratings.

Fig. 1: Comparison of GaN and Si (Image courtesy: www.mouser.com)

